# Raghavan 2021 PDAC — analysis summary

RUN 2 OF 5

## GO / NO-GO GATES

### Gate 1 — integration & continuity → GO

21 samples intermix per-neighbourhood (iLISI 1.0→7.0 of 21) on one continuous component (0 islands). Developmental axis orthogonal to proliferation (|ρ|≤0.15 vs cell cycle) → whole manifold kept.

### Gate 2 — module geography → GO (non-branching)

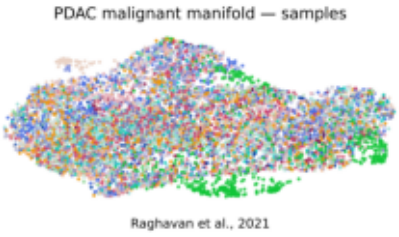
Single Gast/pSom→Crest axis: Gast & pSom co-vary (r=+0.25), both oppose Crest (r=−0.75 / −0.65); Crest an inverse terminal gradient. No Gast-vs-pSom arm split. Crest peaks at the terminus, not the Gast arm → no inversion flag.

### Gate 3 — hub cross-repression → NA

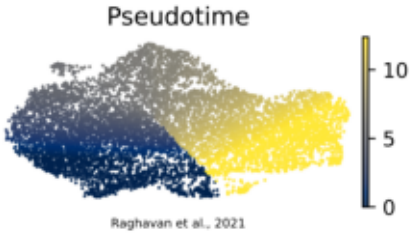
No bifurcation (PPT tree tips share one pSom-high identity; branch-node count unstable 2–4). Branching-only gate does not apply; 04_correlation / 05_modules not produced.

## STEP FIGURES

### Manifold — samples



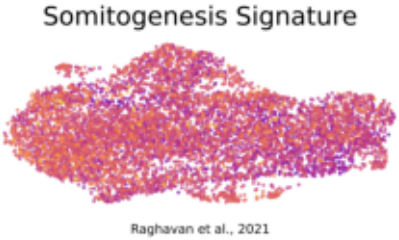

### Pseudotime



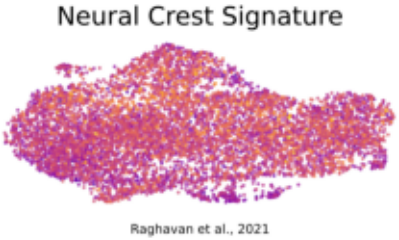

### Gastrulation



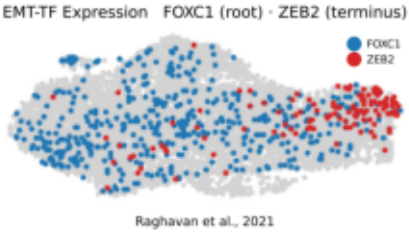

### Somitogenesis




### Neural Crest




### EMT-TF expression




---

**Dataset.**
Metastatic pancreatic ductal adenocarcinoma (Raghavan et al., 2021), Seq-Well — 7,830 malignant cells, 21 samples (mostly liver mets).

**Verdict.**
The embryonic pattern reproduces in collapsed form: one continuous gastrulation/somitogenesis → neural-crest axis (global means Gast −0.011 / pSom +0.072 / Crest −0.024; Linear-pSom/Crest), rooted at the Gast/pSom-high origin, EMT-TFs FOXC1 (root) → ZEB2 (terminus). Non-branching — no competing arms, so Gate 3 is NA. Gate 1 GO / Gate 2 GO / Gate 3 NA.

## Parameter choices

| | | | |
|---|---|---|---|
| **Sample threshold** | geom-mean across largest count gap (120→45) = 73 → 21/23 samples | **Genes / PCA** | overdispersed selection; PCA 50, n_pcs = 20 (43% var) |
| **Integration** | Harmony θ = 10 (mixing plateaus; strongest without collapse) | **UMAP / neighbours** | n_neighbors = 30, spread = 1.3; island removal = radius-graph (0 removed) |
| **Geometry** | principal curve (ElPiGraph), Nodes 20, λ 0.01, μ 50; fold-free (0 forks) | **Root** | derived: Gast/pSom-high, Crest-low tip (Crest is late) |
| **EMT-TFs** | FOXC1 (early) · ZEB2 (late), detection-gated wide panel | | |